Adding and subtracting fractions: answer key

A 1) $6 = 2 \times 3$, $9 = 3 \times 3$: LCM 18 2) $8 = 2 \times 2 \times 2$, $12 = 2 \times 2 \times 3$: LCM 24 3) $10 = 2 \times 5$, $15 = 3 \times 5$: LCM 30 4) $4 = 2 \times 2$, 7 prime: LCM 28

B 1) $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$ 2) $\frac{8}{20} + \frac{5}{20} = \frac{13}{20}$ 3) $\frac{9}{24} + \frac{4}{24} = \frac{13}{24}$ 4) $\frac{5}{30} + \frac{9}{30} = \frac{14}{30} = \frac{7}{15}$

C 1) $\frac{3}{6} - \frac{2}{6} = \frac{1}{6}$ 2) $\frac{15}{20} - \frac{8}{20} = \frac{7}{20}$ 3) $\frac{21}{24} - \frac{4}{24} = \frac{17}{24}$ 4) $\frac{9}{30} - \frac{5}{30} = \frac{4}{30} = \frac{2}{15}$

D Sam added the numerators and also added the denominators. The denominators must be the same before we add, and the denominator does not change. Correct: LCM 12, $\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$.

E 1) $\frac{1}{4} + \frac{1}{6} = \frac{3}{12} + \frac{2}{12} = \frac{5}{12}$ eaten; $\frac{7}{12}$ left ($1 = \frac{12}{12}$). 2) $\frac{2}{5} + \frac{1}{4} = \frac{13}{20}$ spent; $\frac{20}{20} - \frac{13}{20} = \frac{7}{20}$ left.

F 1) $\frac{13}{15}$ 2) $\frac{7}{24}$ 3) $\frac{9}{20}$

G 1) LCM of 12 and 30 = 60 ($12 = 2 \times 2 \times 3$, $30 = 2 \times 3 \times 5$): $\frac{25}{60} + \frac{14}{60} = \frac{39}{60} = \frac{13}{20}$ 2) LCM of 15 and 20 = 60 ($15 = 3 \times 5$, $20 = 2 \times 2 \times 5$): $\frac{44}{60} - \frac{21}{60} = \frac{23}{60}$

★ $\frac{1}{3} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$ (other answers exist).